

THE PLATFORM

One control module. Every power stage.

Power electronics underpins electrification — yet above hobby scale, almost none of it is open. Sealed firmware and proprietary hardware lock out the researchers, educators, and small manufacturers who would otherwise build, study, and improve on it. OpenVVF is the open alternative: one reusable control module plus swappable power stages — motor drive to bidirectional DC-DC, IGBT today, SiC-ready.

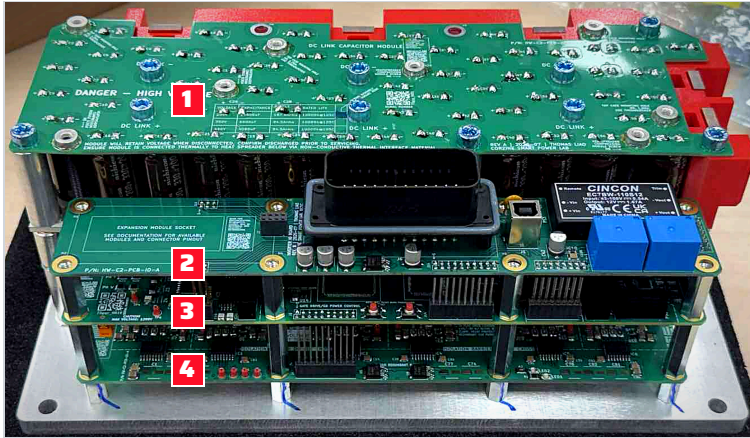


Fig. 1 — The Size 2 stack, top to bottom: capacitor board, IO board, control board, gate-drive board.

COOLING

Three IGBT half-bridge modules conduction-cool to an aluminum baseplate — no liquid loop on Size 2.

SOFTWARE AND WORKFLOW

The same control module runs every chassis: motor self-commissioning (resistance and inductance calibration for PMSM and induction machines), the RTE Studio model-based workflow — fully open-source, no MATLAB required, with a software-in-the-loop pipeline in progress — and the browser-based Telemetry Viewer. The controls program has benchmarked ADRC (active disturbance rejection control) and MPCC (model predictive current control) against the production FOC controller.



Fig. 2 — Ceramic capacitors between the IGBT terminals — the commutation loop measures a few millimeters.

PLATFORM FAMILY

One control module — four power stages

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B / dual CAN FD / expansion · RTE Studio + Telemetry Viewer

Size 1
~50 kW class inverter for light EVs and lab benches
■ **EARLY DESIGN**

Size 2 Inverter
~100 kW · 400 V / 465 A — built and bench-tested
■ **IN VALIDATION**

Size 2 DC-DC
Bidirectional DC-DC for charging, storage, and microgrids
■ **IN PROGRESS**

Size 3
Megawatt class · 900 V / 1,400 A · liquid-cooled; arc-flash containment analysis in progress
■ **IN DESIGN**

EVIDENCE

Bench-tested, on the record

400 V class 150 / 200 / 400 V by DC-link selection	465 A Continuous · 600 A peak (60 s)	10 Published test reports, raw telemetry linked	PMSM + induction Motor self-commissioning, instrument-verified
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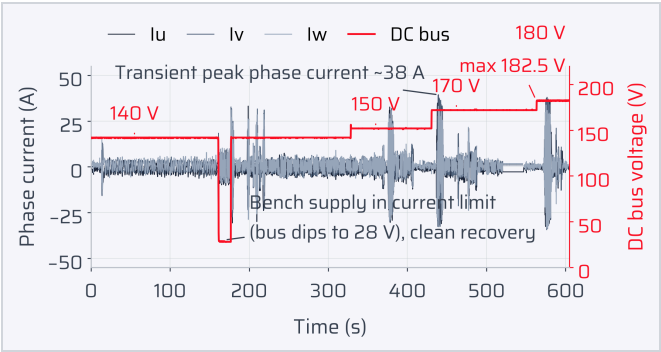


Fig. 3 — 180 V power-stage bring-up: Size 2 chassis (200 V-class DC link) driving a TECO MAX-IE3 10 HP induction motor, bus stepped 140 / 150 / 170 / 180 V. When the bench supply hit its current limit at ~160 s the controller recovered cleanly once the bench-supply current limit was raised — no reset, no fault. Peak phase current ~38 A; maximum bus 182.5 V.

TEST CAMPAIGN	SETUP	KEY RESULT
180 V power-stage bring-up	TECO 10 HP induction motor; bus stepped 140 to 180 V	200 V-class operation confirmed; ~38 A peak; clean fault recovery
1-hour no-load endurance	Size 2 + induction motor, 120 V DC, ~60 Hz	60 min continuous, zero faults; thermal baseline
180 V reversal stress, 20 min	±60 Hz every ~45 s; passive cooling	26 reversals, zero faults; max ~41 A
Self-commissioning accuracy	Inverter estimates vs. LCR meter; PMSM + induction	Resistance within -6.5 to +7.3 %; inductance within +2.4 %

Six further instrumented calibration reports underpin these results — ten reports in all, published with raw telemetry and revised in the open as validation continues.

SAFETY ENGINEERING

Six independent paths to the safe state

The control module is dual-MCU by design. The main MCU (STM32H723ZG) runs FOC motor control, plausibility checks, and fault handling. The safety coprocessor (STM32G474RCTx) independently monitors every safety-critical signal, cross-checks both CAN buses, and runs a challenge/response watchdog — either processor can force the safe state.

Hardware PWM break (TIM1)	<100 ns	Shared driver RESET	<1 μs
1oo2 (one-out-of-two) gate-drive power kill	~10 μs	Independent coprocessor kill	<10 μs
Watchdog-forced main-MCU reset	~100 ms	Passive 3.3 V rail-loss pull-down	passive, no software

ASIL D design target on the top safety goals via ASIL B(D) decomposition — two independent ASIL B channels satisfying an ASIL D goal. ISO 26262 methodology applied as a design input, not a certification claim. Functional-safety roadmap: fault-injection, thermal, and vibration campaigns.

17 hazards identified	>	14 safety goals	>	22 functional safety requirements	>	92 fault-injection tests planned	>	ISO 26262 + IEC 61800-5-2 safe-function mapping
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Safety engineered from the silicon to the sheet metal.

A university lab with a megawatt roadmap

OpenVVVF is built in the Smart Power Lab at UC Santa Cruz. The lab's mission is open power electronics: hardware, firmware, and safety engineering published so others can study, build, and extend it.

Everything the project knows is published as it is learned: schematics and PCB releases, control firmware, hazard and threat analyses, and the raw telemetry behind every test claim. Sponsors and partners make the bench time, fabrication runs, and test campaigns possible.

SUPPORTED BY



TRAJECTORY

■ IN BUILD

Next control-module revision (Gen 7.1)

Hardware revision of the dual-MCU control module.

■ PLANNED

Loaded testing and dyno

Dynamometer campaign on Gen 7.1 takes the platform from no-load validation to rated power.

■ IN PROGRESS

Zero DSR integration

A Size 2 chassis is being readied for our 2018 Zero DSR electric motorcycle.

■ IN PROGRESS

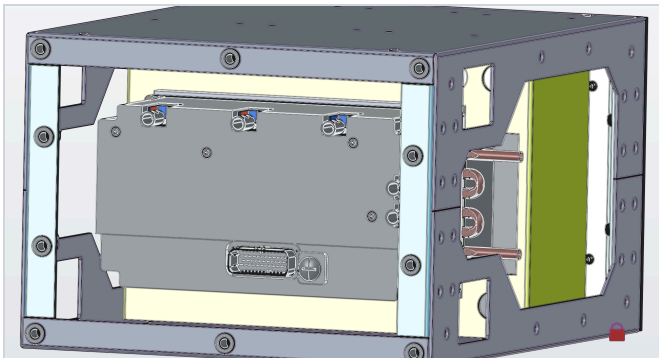
Size 1 + Size 2 DC-DC

~50 kW chassis in early design; a liquid-cooled bidirectional DC-DC variant for charging, storage, and microgrids.

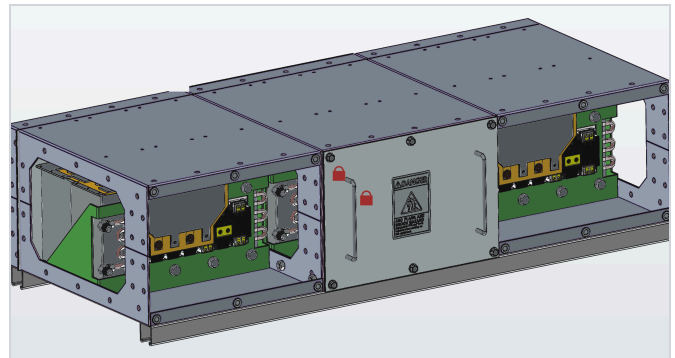
■ IN DESIGN

Size 3 megawatt class

900 V bus, 1,400 A, all-film DC-link; modular liquid-cooled enclosure with arc-flash containment analysis.



Size 2 DC-DC enclosure — design rendering. Reuses the Size 2 power stage and control module in a liquid-cooled enclosure.



Size 3 megawatt-class enclosure — design rendering. Modular liquid-cooled bays, arc-flash containment analysis.

Every schematic, line of firmware, safety analysis, and test trace is public at docs.openvvvf.org.

OPEN HARDWARE · OPEN FIRMWARE · OPEN EVIDENCE